AI / ENTERPRISE SAAS • CASE STUDY

TwinForce — The Autonomous Enterprise

Building an enterprise-grade AI digital twin platform that learns employee communication patterns and automates repetitive knowledge work at scale.

2024–2025	Enterprise SaaS	Global Clients	Multi-tenant

THE CHALLENGE

Enterprises lose thousands of hours to repetitive knowledge work

Large organizations struggle with a core problem: highly skilled employees spend the majority of their time on repetitive tasks — drafting emails, summarizing meetings, answering the same internal questions repeatedly. TwinForce was built to solve this by creating an AI digital twin of each employee that learns their communication style, institutional knowledge, and workflows — then handles the repetitive layer autonomously.

40%	10x	300+	3
Time saved per employee	Faster meeting follow-ups	API endpoints built	LLMs orchestrated

WHAT WE BUILT

Four core systems working together

AI Twin Engine

Each employee gets a personal AI twin trained on their emails, Slack messages, and documents. It learns tone, vocabulary, and decision-making patterns to respond authentically.

Meeting Intelligence

Whisper API transcribes meetings in real time. The twin auto-generates summaries, extracts action items, and drafts follow-up emails — all before the call ends.

RAG Knowledge Base

LlamaIndex + Qdrant vector DB powers semantic search across all company documents, Notion pages, and Confluence wikis — returning answers with source citations in under 2 seconds.

Email Intelligence

The twin reads incoming threads, drafts replies in the employee's voice, and routes for one-click approval — cutting email handling time by over 60%.

TECH STACK

Enterprise-grade architecture

Next.js 15	TypeScript	Hono.js + Bun	FastAPI Python
LangGraph	LlamaIndex	GPT-4o	Claude 3.5 Sonnet
Whisper API	ElevenLabs	PostgreSQL + Supabase	Qdrant (vector DB)

Redis + NATS	Docker + Kubernetes

DELIVERY TIMELINE

How we shipped it

Phase 1 — Foundation	PostgreSQL schema (20+ tables), multi-tenant RLS policies, JWT auth, RBAC roles, Supabase setup, Docker Compose for all services.
Phase 2 — AI Layer	Twin training pipeline (LangGraph), RAG ingestion (LlamaIndex to Qdrant), Whisper transcription, ElevenLabs voice synthesis, email intelligence drafting.
Phase 3 — API & Integrations	300+ REST endpoints (Hono.js), Stripe billing, Zoom/Google/Microsoft OAuth, NATS event streaming, audit logs, webhook system.
Phase 4 — Production	Kubernetes Helm charts, Terraform (AWS), GitHub Actions CI/CD, Prometheus + Grafana monitoring, Sentry error tracking.

RESULTS

Impact delivered

- ✓ Employees reclaim 40% of their week by offloading routine emails, meeting summaries, and internal Q&A; to their AI twin.
- ✓ Meeting follow-ups that previously took 45 minutes are generated and ready for approval within 2 minutes of the call ending.
- ✓ Knowledge retrieval across 10,000+ company documents in under 2 seconds with source citations — eliminating knowledge bottlenecks.
- ✓ Full multi-tenant SaaS with Stripe billing (Starter $30/mo to Enterprise $50/mo), ready for 100+ enterprise clients from day one.

> *"TwinForce is what we imagined AI should actually do in the enterprise — not just answer questions, but genuinely take work off your plate in your own voice and style."*
>
> **— Enterprise client, Series B SaaS company**

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